

GESER DUGAROV, Ph.D. | Software Engineer | Distributed Data Systems & AI Data Infrastructure

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SUMMARY

Software Engineer building data lakehouse and AI data infrastructure for petabyte-scale data, currently a distributed cache and an incrementally updatable vector index targeting tens of billions of vectors. Open-source contributor to Apache Hudi (50 [merged PRs](#)) and Lance (3 [merged PRs](#)).

Building open-source tooling for agentic software development – issue-driven orchestrators and develop-review loops that turn coding-agent CLIs into reproducible engineering pipelines. Extensive experience in research and data analysis; PhD. Strong interest in big data, distributed systems, performance engineering, and product-focused backend development.

TECHNICAL SKILLS

Java, Python, Rust (some experience), Hudi / Lance, Spark / Flink / Ray, vector search / ANN indexing, distributed systems, LLM agent orchestration, multi-agent workflows

WORK EXPERIENCE

May 2023 – Present **Software Engineer / Big Data Engineer,**
(3+ yrs) [Huawei](#)

Development of core functionality within Huawei Cloud for big data processing on enterprise-scale clusters.

- Designed and implemented a distributed cache for Lance, pairing an in-memory L1 with a persistent L2 on per-worker NVMe and prewarming IVF partitions to avoid repeated object-storage reads.
- Building a distributed, incrementally updatable vector index for datasets containing tens of billions of vectors using query-adaptive ANN search.
- Optimized Flink stream writing, halving processing time.
- Reduced the platform's configuration burden by making sane Hudi defaults automatic for common scenarios.
- Implemented partition-level TTL, enabling customers to automate cloud storage cost management with coarse granularity.

Jan 2024 – Present **Apache Hudi Contributor,**
(2.5+ yrs) **The Apache Software Foundation**

[Apache Hudi](#) is a data lakehouse platform that brings database functionality to data lakes and enables incremental processing for low-latency analytics.

- Optimized record serialization and deserialization in Flink stream writes to Hudi tables, improving processing speed by 30% and halving memory usage ([design doc](#), [main changes](#)). Released in [Hudi 1.0.2](#).
- Implemented four targeted optimizations ([\[1\]](#), [\[2\]](#), [\[3\]](#), [\[4\]](#)), improving processing speed by 10% and reducing garbage-collection overhead by 30%. Released in [Hudi 1.0.1](#).
- Author of RFC-98 for Spark DataSource V2 read support; the design and copy-on-write implementation are under review.

- Authored 50 [merged pull requests](#).

May 2026 – Present **Open-source contributor,
Lance ecosystem**

Rust contributions to the [Lance columnar format](#) and its distributed tooling. Lance is an open-source lakehouse format that unifies fast multimodal data access, indexing, and AI training.

- Authored three [merged pull requests](#) in the vector-index build path, reducing memory use and cleaning up temporary resources in long-running index jobs.

Apr 2026 – Present **Open-source: Agentic Development Tooling,
Personal projects**

Self-directed work on AI-powered developer tooling: multi-agent implementation and review workflows with human-in-the-loop handoff, reproducible execution, monitoring, and cost tracking.

- [agent-orchestrator](#) – Python-based, GitHub issue-driven workflow runner. Decomposes an issue, runs an implementer agent in an isolated git worktree, opens a PR, runs an independent reviewer agent, notifies a human when ready to merge. State persists in the issue itself. Eight releases from v0.1.0 (May 2026) to v0.8.0 (Jul 2026). Operators pause/continue controls, PostgreSQL-backed usage and cost analytics, and an agent-trajectory viewer. Operational telemetry since May 2026: 5,100+ agent runs across 12 repositories, including 1,500+ independent reviewer runs and an estimated \$11k+ in API-equivalent LLM usage cost.
- [agent-develop-review-loop](#) – shell-only develop-review loop with HITL handoff, resumable runs, and cost telemetry parsed from agent usage metadata.

Feb 2022 – May 2023 **Software Engineer / ML Engineer,
(1+ yr) Digital Research (computer vision startup)**

- Designed an event-driven architecture and implemented the image-processing backend in Python for a truck-monitoring system handling ~20,000 images/day; in production, the system reduced fleet idle time by 12%.
- Built a customer-facing web UI featuring reports and data visualizations. Also developed an internal web UI for system monitoring.

EDUCATION

PhD, Geophysics, Trofimuk Institute of Petroleum Geology and Geophysics SB RAS

MSc, Computational and Applied Mathematics, Novosibirsk State University

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